

Evidence-Based Professional Development

Research Brief for Josh

Level 1 IT Support · Dubbo Christian School

What the learning science actually says — and what it means for your Thursday/Friday PD blocks, your CompTIA A+ progress, your DCSPrep app, and your growth toward Level 2.

Compiled 08 May 2026 · Parts A – I · Peer-reviewed sources throughout

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Executive Summary

What the evidence agrees on. The case for distributing practice over time (spacing), testing yourself from memory rather than re-reading (retrieval practice), and writing/explaining material in your own words (generation) is among the most replicated in cognitive psychology, with meta-analytic effect sizes in the $d = 0.40$ – 0.60 range for applied settings. These are not contested ideas — they are the bedrock of evidence-based self-directed learning for adult professionals.

What is often misunderstood. Feeling productive is not the same as learning. Blocked, sequential study of a single topic feels smooth and fast; it also produces rapid forgetting. The opposite — interleaving, spacing, and self-testing — feels harder and slower during the session, but produces substantially better retention after a delay. This is Bjork's 'desirable difficulties' concept in a nutshell, and it is directly relevant to how you structure your Thursday/Friday blocks.

Where caution is warranted. Most laboratory studies use undergraduates in single-session designs, which limits direct extrapolation to a 31-year-old support tech with ADHD, two structured PD days, and live interruptions. The effect sizes in applied, classroom, and workplace studies are smaller but remain positive. Learning styles (VAK matching) are not supported. The 25/5 Pomodoro number has no special evidence base — block lengths should reflect your actual task structure and energy, not a catchy timer.

For your specific context. Your Thursday/Friday blocks total approximately 13 hours 20 minutes per week — a substantial resource. The research consistently shows that how you use that time matters far more than how much of it you have. An hour of retrieval practice, hands-on troubleshooting, and writing outperforms three hours of passive video re-watching. Your DCSPrep app, ticket-note writing, runbooks, and scenario labs are not supplementary to your study — they are some of the most evidence-aligned activities you could do.

On sustainability. Burnout in self-directed adult learners is linked to perceived lack of progress, over-prescription, and loss of autonomy. The antidote is a visible evidence pack, a flexible schedule that degrades gracefully under interruption, and a commitment to recovery breaks as non-negotiable — not as wasted time. The sleep between Thursday and Friday is a legitimate learning tool: overnight consolidation is well-supported, and Friday's blocks genuinely build on Thursday's foundation.

On ADHD as a design constraint. Context-switching costs are measurably higher for adults with ADHD than neurotypical controls (Sergeant et al., 2002; Mueller et al., 2017). The implication is protective routine, not restriction: a clear written re-entry ritual after interruptions, session anchors (a 3-sentence intention written at the top of the hour), and activity types that match current energy. Hyperfocus is real and can be productive; the risk is neglecting transitions and breaks. Build your app to notice, not to nag.

One-sentence version. Space it out, test yourself, write and explain, take real breaks, use your app and ticket notes as study, and trust the overnight sleep — this is what the evidence says.

Foundational Learning Science

Eight evidence areas translated into practical implications for a DCS IT support worker

A1 — Spacing Effect & Distributed Practice

● **STRONG**

The spacing effect — the finding that distributing practice sessions across time produces better long-term retention than massing the same amount of practice into a single session — is one of the most replicated findings in cognitive psychology. A 2024 classroom meta-analysis (Burgess & Jilbert, Behavioral Sciences) found a moderate pooled effect of $d = 0.54$ favouring distributed over massed practice across 31 effect sizes, with larger effects at longer retention intervals.

For adult professional learners, the optimal inter-session interval is topic- and expertise-dependent. The principle is 'expand gradually': new, difficult material benefits from short initial gaps (hours to 1–2 days), while consolidating material tolerates and benefits from longer gaps (1–2 weeks). With only two structured PD days per week, every Friday session becomes the first spacing event for Thursday's new material — a structurally desirable gap of ~16 hours plus overnight sleep. The five-day gap between Friday and the following Thursday provides the second, longer spacing interval.

Caution: spacing effects are somewhat attenuated for highly complex procedural knowledge (Ruitenburg et al., 2025, Applied Cognitive Psychology; Donovan & Radosevich, 1999). For multi-step troubleshooting flows (not just ports or definitions), you may need more massed initial exposure before spacing kicks in.

Key citations: Cepeda et al. (2006), Psychological Bulletin; Burgess & Jilbert (2024), Behavioral Sciences; Ruitenburg et al. (2025), Applied Cognitive Psychology; Donovan & Radosevich (1999), Journal of Applied Psychology.

Practical Implication

So Josh could: treat Thursday as intake + first practice, and Friday as the first spaced retrieval event for Thursday material. Use your DCSPrep flashcard deck with FSRS scheduling so the algorithm spaces each card's next review based on your actual recall performance — not a fixed calendar.

A2 — Retrieval Practice & the Testing Effect

● **STRONG**

Actively retrieving information from memory (recall, cued recall, self-quizzing) produces substantially better long-term retention than restudy of the same material. The median testing-effect size across 159 effect sizes in Rowland's (2014) meta-analysis was $g = 0.55$. The benefit is larger when the retention interval is at least one day — precisely the Thursday-to-Friday gap you already have.

Free recall (writing everything you remember without prompts) and cued recall (answering a specific question from memory) both outperform recognition (multiple-choice) for durable retention, though recognition still beats restudy. The CompTIA practice exam format (multiple-choice) is therefore valuable for exam preparation but should be supplemented with open recall — writing out 'what does this error state mean?' from scratch.

Retrieval practice works for procedural knowledge too. Kromann et al. (2009) found retrieval practice improved application of clinical procedures. Parker & Roessger (2020, IJLE) specifically note that IT certification learners using self-made or proprietary practice exams without instructor direction can achieve strong testing-effect benefits — provided they retrieve before checking answers, not after.

Feedback timing: immediate corrective feedback is better than no feedback, but delayed feedback (end of quiz, not per item) may produce slightly better long-term retention due to the generation of error-related learning during the delay. In practice, immediate feedback in your DCSPrep quiz mode is fine and far better than no feedback.

Key citations: Roediger & Karpicke (2006), Psychological Science; Rowland (2014), Psychological Bulletin; Parker & Roessger (2020), IJLE; Kromann et al. (2009), Medical Education.

Practical Implication

So Josh could: at the start of each Friday Block 1, do a 10–15 minute brain-dump of what you studied Thursday before opening any notes. Write port numbers, troubleshooting steps, and concepts from memory. Then check — what was missing is your highest-priority review topic for the day.

A3 — Interleaving vs Blocked Practice

● MODERATE

Interleaving — mixing different topics or problem types within a session rather than blocking them sequentially — improves long-term retention and transfer compared to blocked practice, particularly when learners must discriminate between similar concepts (Rohrer, 2012; Firth et al., 2021 systematic review, Review of Education). The effect is well-established for mathematics and concept discrimination; evidence for applied procedural skills is moderate but growing.

When NOT to interleave: early acquisition of a completely new concept benefits from some initial blocked exposure to build the schema before interleaving is introduced. If you are encountering IP subnetting for the first time, block it for one full session before mixing it with other networking topics. The expertise reversal effect (Kalyuga et al., 2003) also applies here: as you become more expert, purely blocked re-study provides diminishing returns.

For your situation, multiple sources (CompTIA Core 2, Cisco, La Trobe, DCSPrep) covering overlapping topics (networking, security, OS) creates a natural interleaving opportunity — but only if you consciously revisit the topic across sources rather than finishing one source entirely before starting another. This 'rich interleaving' produces robust transfer; random source-hopping without a topic anchor produces shallow coverage.

Key citations: Rohrer (2012), Educational Psychology Review; Firth et al. (2021), Review of Education; Bjork & Bjork (2020), Desirable Difficulties in Theory and Practice.

Practical Implication

So Josh could: after your first full pass on a CompTIA topic, resist the urge to immediately move to the next video. Instead, open a DCSPrep scenario or a Cisco question on the same topic — that interleaving across formats is where the transfer is built.

A4 — Desirable Difficulties (Bjork)

● STRONG

Bjork's 'desirable difficulties' framework (Bjork & Bjork, 1992; 2020) identifies conditions that degrade short-term performance but enhance long-term learning: spacing, interleaving, retrieval practice, generation, and reduced feedback. The counterintuitive core finding is that conditions that feel productive (blocked practice, re-reading, immediate feedback) are often the least effective for lasting retention.

For ADHD learners specifically, desirable difficulties present a design challenge: the effort associated with spacing and interleaving can trigger avoidance, and hyperfocus tendencies mean it is tempting to stay in the smooth, fast, massed mode. An app that names the difficulty ('this feels harder because it's spaced — that's correct') is protective rather than nagging.

Interleaving showed 30–40% better outcomes than blocked practice in mathematics studies (Taylor & Rohrer, 2010; 2013), with the advantage growing when learners must select the solution method — exactly the skill relevant to IT troubleshooting where problem type is not pre-announced.

Key citations: Bjork & Bjork (2020); Taylor & Rohrer (2010, 2013), Journal of Applied Research; Smith & Weinstein (2016).

Practical Implication

So Josh could: when a quiz or scenario feels annoyingly hard, that is probably a signal it is working — not a signal to switch to something easier. Use your PD log to note 'this was hard today', and check in 48 hours whether recall improved.

A5 — Multimodal Learning & Dual Coding (not Learning Styles)

● MODERATE

Learning styles (VAK) are not supported. Pashler et al. (2008, *Psychological Science in the Public Interest*) reviewed the evidence and found no reliable support for the hypothesis that matching instructional format to a learner's preferred modality improves learning outcomes. This remains the scientific consensus.

What IS supported is dual coding (Paivio, 1971; Mayer, 2009): combining verbal and visual representations of the same information (e.g., a network diagram alongside a written explanation of how packets route) activates complementary encoding pathways, producing better retention than either alone. This is not about learning style — it works for virtually everyone for the same material.

The practical implication for modality variation is engagement and fatigue management, not neurological matching. Rotating between video, flashcards, hands-on lab, and writing across a 155-minute block prevents sustained attention fatigue from a single format — a valid reason to vary activities that is distinct from (and better supported than) learning styles claims.

Key citations: Pashler et al. (2008), Psychological Science in the Public Interest; Paivio (1971); Mayer (2009), Multimedia Learning (2nd ed.).

Practical Implication

So Josh could: draw your own network diagram while watching the Messer video, rather than watching twice. The act of sketching the topology while hearing the explanation activates dual coding and forces active engagement — more effective than a second passive viewing.

A6 — Generation Effect, Self-Explanation & the Protege Effect

● STRONG

The generation effect (Slamecka & Graf, 1978; Bertsch et al., 2007 meta-analysis, 86 studies, medium-to-large effect) shows that information you produce yourself is remembered better than information you read. Writing your own ticket template, runbook, or DCSPrep scenario is not just output — it is a high-quality study act.

Self-explanation (explaining solved steps in your own words without looking at the source) shows a weighted effect size of $g = 0.55$ across 64 studies (meta-analysis cited in Fiorella & Mayer, 2016). The effect is strongest when explanations are generated *without* access to the source — the gaps you discover are precisely your knowledge weaknesses.

The protege effect (explaining to a real or imagined novice) further strengthens retention by activating metacognitive monitoring. When you write a DCSPrep scenario, imagine explaining it to a first-year tech — this framing consistently improves the depth of the explanation produced and the recall of the material used.

For your context: writing ticket notes that explain what the problem was, what you tried, and why the fix worked is a generation + self-explanation act. It is not overhead — it is some of the most productive time you

spend. The same applies to PD log entries and runbooks written from memory before checking documentation.

Key citations: Slamecka & Graf (1978); Bertsch et al. (2007); Fiorella & Mayer (2016), Educational Psychology Review; Dunlosky et al. (2013), PSPI.

Practical Implication

So Josh could: write your first draft of any runbook or ticket template from memory before opening documentation. The gaps that appear are your study agenda. Writing the explanation IS the study — not the preparation for study.

A7 — Cognitive Load Theory (Sweller)

● **STRONG**

Cognitive load theory (CLT; Sweller, 1988; van Merriënboer & Sweller, 2010) proposes that working memory is severely limited, and that instructional design should minimise extraneous load (caused by poor presentation) while managing intrinsic load (caused by topic complexity) and directing resources to germane load (schema construction).

Worked examples reduce extraneous load for novices by providing solution steps to study before attempting problems independently. Professor Messer's videos are, in CLT terms, a form of worked example — explanatory demos that build schema before you encounter the problem raw.

The expertise reversal effect (Kalyuga et al., 2003) means that as you move from novice to intermediate, the same worked-example format becomes less useful and can actually impede learning. For CompTIA topics you have already studied (Core 1 material), problem-solving scenarios outperform re-watching the video. For new Core 2 topics (Windows Settings, scripting), video first remains appropriate.

Chunking and simple-to-complex sequencing reduce intrinsic load: break subnetting into CIDR notation, then subnet masks, then VLSM — not all at once. DCSPrep scenarios that combine multiple unknowns should be reserved for later stages.

Key citations: Sweller (1988, 1994, 2024); Kalyuga et al. (2003); van Merriënboer & Sweller (2010), Medical Education.

Practical Implication

So Josh could: for any Core 2 topic, watch the Messer video first (worked example, reduces load). For any Core 1 topic you want to revise, skip the video and go straight to a DCSPrep scenario or practice question — the expertise reversal effect predicts better return on that effort.

A8 — Metacognitive Monitoring & Judgements of Learning

● **STRONG**

Learners systematically overestimate how much they have retained from a study session (Koriat & Bjork, 2005; Dunlosky & Metcalfe, 2009). Re-reading or re-watching produces an 'illusion of knowing' — familiarity is mistaken for recall ability. This is especially acute for ADHD learners where variability in attention during encoding creates unreliable self-assessment.

Self-testing rituals correct for this. When you close the Messer video and immediately write down the key points from memory, the gaps between what you thought you learned and what you can actually retrieve become visible. That discrepancy is more actionable than any 'confidence rating' slider.

For self-directed PD where no instructor checks the work, the PD log functions as a metacognitive scaffold: 'What did I set out to learn? What can I now explain without notes? Where are the gaps?' Three questions, two minutes. Without this ritual, the week fills with activity that feels productive but produces shallow encoding.

Key citations: Koriat & Bjork (2005); Dunlosky & Metcalfe (2009); Dunlosky et al. (2013), PSPI.

Practical Implication

So Josh could: at the end of every study block — even if it's only 20 minutes left after an interruption — write three sentences in your PD log: what you worked on, what you can now explain without notes, and one gap. This takes less than 2 minutes and prevents the illusion of knowing from accumulating.

Recovery & Sustainable Practice

Design constraints on the schedule — not the primary topic, but non-negotiable

This section treats recovery as a hard constraint on the schedule. Every finding below ends in a schedule implication. None of this is clinical advice.

B1 — Microbreaks During Cognitively Demanding Work

Albulescu et al. (2022, PLOS ONE) meta-analysed 22 studies (N=2,335) and found small but significant effects of microbreaks (<10 min) on vigour ($d=0.36$) and fatigue reduction ($d=0.35$). Critically, for high-cognitive-demand tasks, **longer breaks are needed for performance recovery** — the micro-break effect on performance was non-significant. This means a 3-minute walk to the water fountain is useful for mood but may not fully restore deep problem-solving capacity. Within a 155-minute block, build in a deliberate 5–8 minute movement break at the midpoint (around minute 75–80). Albulescu's meta-regression: the longer the break, the greater the performance boost.

Schedule Implication

Schedule implication: treat the midpoint break in Block 1 and Block 3 as mandatory, not optional.

B2 — Walking, Movement & BDNF

Moderate aerobic exercise elevates Brain-Derived Neurotrophic Factor (BDNF), which supports synaptic plasticity and memory consolidation (Erickson et al., 2011, PNAS). Short walks between cognitive blocks are supported by attention restoration theory (Kaplan & Kaplan, 1989): natural environments (even a 10-minute outdoor walk) restore directed attention capacity. Your recess and lunch walks are therefore high-leverage recovery activities — not a distraction from study. The evidence for walking specifically enhancing consolidation of the just-finished block is suggestive rather than definitive, but walking outdoors during breaks is consistently among the most protective recovery activities identified across studies.

Schedule Implication

Schedule implication: walk outside during at least one break each day — the evidence for this is stronger than any specific block structure recommendation.

B3 — Eating & the Post-Prandial Dip

The post-lunch dip — a circadian-driven secondary sleep propensity peak occurring roughly between 1:00 and 4:00 pm — is well-documented (Monk, 2005; Monk et al., 1996). It occurs even without a meal and is worsened by high-carbohydrate foods. Your Block 3 begins at 1:45 pm, directly in the dip window. High-carbohydrate lunches substantially worsen performance in this window. Bright light exposure in the early afternoon (Zhou et al., 2021, Frontiers in Public Health) partially counteracts the dip. A light, protein-moderate lunch, outdoor light exposure during the lunch break, and starting Block 3 with a hands-on or social activity (not passive video) all help.

Schedule Implication

Schedule implication: eat a light lunch, go outside during the 40-minute break, and begin Block 3 with an active task (DCSPrep scenario, ticket template writing, hands-on troubleshooting) rather than passive video.

B4 — Chronotype & Time-of-Day Effects

Sustained attention and working memory performance peak during the late morning to midday for most adults (intermediate chronotypes), with a secondary peak in the late afternoon (Dijk & Czeisler, 1995; PMC 2023 review). Morning chronotypes show the post-lunch dip most strongly. For an 8:30am start: Block 1 morning hours are highest-alertness for most people — optimal for new encoding. Block 2 (11:25/11:35 am) remains good. Block 3 (1:45 pm) coincides with the dip and is best used for application, retrieval, and building tasks rather than new complex intake.

Schedule Implication

Schedule implication: new CompTIA video content and hardest abstract material belong in Block 1. Block 3 is for DCSPrep building, ticket writing, retrieval practice, and troubleshooting scenarios.

B5 — Sleep Consolidation Between Thursday & Friday

Sleep supports memory consolidation via hippocampal replay and cortical transfer (Walker, 2017). Guttesen et al. (2025, Cognition) confirmed that sleep improves retention of both item and associative memories, irrespective of whether sleep followed quickly or after a prolonged waking interval. The night between Thursday and Friday is therefore a legitimate part of your learning architecture. What the research does NOT support: the expectation that everything from Thursday will be perfectly retained Friday morning. Retrieval practice at the start of Friday (the brain-dump ritual from A2) is still essential — it surfaces what sleep consolidated and what still needs work.

Schedule Implication

Schedule implication: begin Friday Block 1 with a 10–15 minute free recall of Thursday's material before reviewing any notes. Trust the overnight sleep, then test it.

B6 — ADHD as a Design Constraint

Adults with ADHD show measurably higher context-switching costs than neurotypical controls (Mueller et al., 2017; Sergeant et al., 2002), particularly on attentional set-shifting (PMC 2018). Hyperfocus is a real phenomenon affecting ~68% of adults with ADHD (PMC 2024) and correlates with missed transitions and neglected breaks (40% of hyperfocus episodes). Context-switching costs are worsened by unpredictability — thus, a reliable daily structure reduces cognitive overhead even if the tasks within it vary. The app should use **predictive not reactive** structures: 'at minute 80, this is your movement break' is less disruptive than a surprise ping.

Schedule Implication

Schedule implication: write a 2–3 sentence intention at the start of every block, set a midpoint physical timer, and build the re-entry ritual ('three-line PD log, restate the goal') into the app's interruption-resume flow.

B7 — Burnout Signals & Protective Factors

Self-directed adult learner burnout is predicted by perceived lack of autonomy, invisible progress, and unsustainable pace (Ryan & Deci, 2000, self-determination theory; Schaufeli et al., 2009). Protective factors: visible evidence of progress (your Evidence Pack is structurally protective), genuine choice in activity sequence, and a pre-defined 'deload' protocol when the week has been heavily interrupted. A week of mostly operational support should produce lighter PD expectations, not shame.

Schedule Implication

Schedule implication: the app should track your weekly PD hour total and suggest a 'consolidation week' mode (lighter retrieval practice only, no new intake) if two or more full blocks have been lost to operational work in the same week.

Structuring a Single Study Block

Evidence-grounded internal session shapes for 90, 100, 145 and 155-minute blocks

The 'ultradian / BRAC' literature (Kleitman) suggesting a strict 90-minute biological work cycle is **not well-supported** as a prescription for cognitive work. The 90-minute figure is derived from REM sleep cycles and has not been robustly established as a general attention rhythm. However, practitioner experience and fatigue research do suggest that sustained high-cognition work benefits from a deliberate activity change and brief movement within any block longer than 90 minutes. The shapes below are grounded in cognitive load theory, the testing effect, and fatigue management evidence — not in the BRAC cycle claim.

90-Minute Block (Friday Block 2, 11:35–1:05pm)

Time	Activity Type	Evidence Rationale
0–5 min	Orientation	Write block intention (3 sentences). Brief retrieval warm-up: what are the 3 key things from the last session? This primes retrieval networks (testing effect, Roediger & Karpicke, 2006).
5–30 min	Active Intake / Retrieval	Whichever is due: new Messer video segment OR retrieval practice on prior topic. If new: one section only (not multiple topics). If retrieval: free recall then flashcards.
30–55 min	Application	DCSPrep scenario, hands-on troubleshooting, or scenario-based quiz on the just-studied topic. Cognitive load theory: application after worked example consolidates schema.
55–80 min	Writing / Building	Ticket template, runbook, or DCSPrep content on the session topic. Generation effect: producing the explanation deepens encoding.
80–88 min	Movement Break	Stand, walk around the office, water. Brief physical reset. Albulescu et al. (2022): microbreaks reduce fatigue $d=0.35$.
88–90 min	PD Log Closure	Three-sentence log entry: what was studied, what can be explained without notes, one gap. Metacognitive closure prevents illusion of knowing.

100-Minute Block (Thursday Block 2, 11:25–1:05pm)

Time	Activity Type	Evidence Rationale
0–5 min	Orientation + Retrieval Warm-up	Block intention (written). 3-question free recall of previous block material.
5–35 min	Primary Activity	New video content (Block 1 residue) OR retrieval practice (if Block 1 was heavy intake). 30 minutes is sufficient for 2–3 Messer sub-sections.
35–65 min	Application or Building	DCSPrep scenario, IT lab exercise, runbook draft, or DCSPrep content authoring on the primary topic.
65–70 min	Microbreak	Movement. If outside is accessible, a brief 5-minute outdoor walk is optimal.
70–95 min	Interleaved Retrieval	Flashcard review on a DIFFERENT topic from the primary activity (deliberate interleaving). 15–20 cards from DCSPrep SRS deck.
95–100 min	Log + Handoff	PD log entry. Write the first sentence of the next session's goal so re-entry is easier (reduces context-switch cost on return).

145-Minute Block (Thursday Block 1, 8:30–10:55am)

Time	Activity Type	Evidence Rationale
0–8 min	Orientation + Full Recall	Block intention. Extended brain-dump: 5 minutes writing everything remembered from the previous PD day before any review. This is the highest-leverage retrieval moment of the week.
8–45 min	New Intake — Segment A	New Messer content or complex new topic (hardest material). Morning alertness is at peak. Dual-code: sketch diagrams while watching.
45–75 min	Application — Segment A	Hands-on, DCSPrep scenario, or practice questions on the just-watched content. Not more video.
75–83 min	Movement Break	Mandatory midpoint break — stand, walk, ideally brief outdoor exposure. Research: longer break = greater performance boost (Albulescu meta-regression).
83–110 min	New Intake — Segment B OR Retrieval	If energy is high: second new topic segment. If energy is lower: switch to retrieval practice / flashcard deck on a prior topic. Read your own energy.
110–135 min	Building / Writing	DCSPrep content creation, runbook drafting, ticket template writing. Generation effect: this is study, not admin.
135–143 min	Retrieval Closure	5-card targeted retrieval on any topic that felt shaky during the block.
143–145 min	PD Log	3-sentence close. One gap identified.

155-Minute Block (Thursday/Friday Block 3, 1:45–4:20pm; Friday Block 1, 8:30–11:05am)

Time	Activity Type	Evidence Rationale
0–8 min	Orientation	Written intention. For Block 3: acknowledge the post-prandial dip is real — begin with active/hands-on, not passive video. For Friday Block 1: extended recall brain-dump from Thursday.
8–40 min	Primary Activity — Segment A	Active application, troubleshooting scenario, DCSPrep building, or retrieval practice. Avoid new passive video in Block 3 afternoon hours.
40–75 min	Primary Activity — Segment B	Continue application, or shift to a complementary topic. Building DCSPrep app features is appropriate here — engaged, creative, low new-intake burden.
75–85 min	Mandatory Internal Break	10-minute break is non-negotiable in a 155-minute block. Move, eat a small snack if needed, go outside briefly. Albulescu: for high-cognition tasks, performance recovery needs >10 min — this is the minimum.
85–115 min	Retrieval + Flashcards	Spaced repetition deck review (DCSPrep/Anki FSRS). 15–20 cards across multiple topics. This is the interleaving interval — mix topics rather than drilling one subject.
115–140 min	Writing / Evidence Pack	Ticket template, escalation note, Evidence Pack log entry, manager-safe documentation. Meaningful output that serves both study and professional record.
140–150 min	Soft Wind-Down	Low-demand review: skim notes from the block, correct any obvious gaps in DCSPrep content.
150–155 min	PD Log + Next Session Seed	3-sentence log. Write the first sentence of next session's goal.

The Between-Block Break

30-minute Recess and 40-minute Lunch as structured recovery — not wasted time

The recess and lunch breaks are recovery intervals, not study time. Attempting to use them for additional content intake is counterproductive: it prevents consolidation of the just-finished block, denies the cognitive reset that makes the next block productive, and compounds fatigue across a three-block day.

What supports consolidation during breaks:

- Quiet wakeful rest (even 5–10 minutes of mind-wandering) — replay/consolidation continues during wake rest (Dewar et al., 2012)
- Walking outdoors — attention restoration (Kaplan & Kaplan, 1989), BDNF support (Erickson et al., 2011)
- Light social conversation — low-load, positive affect supports memory consolidation
- Eating a protein-moderate meal — avoids the glucose spike-and-dip worsening Block 3 fatigue
- Sunlight exposure — counteracts post-prandial dip (Zhou et al., 2021)
- Hydration — even mild dehydration impairs working memory and attention

What interferes with consolidation:

- More cognitive content on the same topic (prevents consolidation replay from completing)
- Novel verbal input (podcasts, videos, complex reading) — new encoding competes with consolidation
- High-stress social interaction — cortisol elevation impairs hippocampal consolidation
- Screens with notifications — each notification triggers a small context switch with attention residue
- Extended social media scrolling — mild working memory depletion from rapid context-shifting

Are 30 and 40 minutes sufficient?

Albulescu et al. (2022) found that for high-cognitive-demand tasks, microbreaks under 10 minutes are insufficient for full performance recovery. Thirty minutes of physical movement and food is on the boundary of adequacy. Forty minutes at lunch — if the lunch break includes outdoor walking and a light protein-rich meal — is more reliably restorative. The key variable is what you DO during those 30/40 minutes, not the duration alone.

Practical break templates:

Recess Template A — The Reset Walk

Leave the building immediately. Walk for 15–20 minutes outdoors, no phone, no podcasts. Return, eat. No study content until Block 2 begins.

Recess Template B — The Social Reset

Eat with colleagues, light conversation. Briefly step outside for 5–10 minutes of natural light. Avoid discussing IT problems — this keeps the conversation low-cognitive-load.

Lunch Template A — Full Decompress

Walk outdoors 20 minutes (ideally green space). Eat a light protein-moderate meal. 5 minutes of quiet rest or light prayer/journaling. No content intake.

Lunch Template B — Movement + Prep

Walk 15 minutes. Eat. In the final 5 minutes, write your Block 3 intention (3 sentences only). This brief future-orientation is low-load and supports clean Block 3 entry.

Sequencing Across a Study Day

Which PD belongs in Block 1, 2, and 3 — and why

Block	Best PD Types	Evidence Rationale
Block 1 (Morning, Fresh) 8:30am onwards	New CompTIA/Messer intake; hardest abstract material (subnetting, security protocols); extended brain-dump recall of previous session; anything requiring high working memory and sustained attention.	Peak alertness and working memory. Primacy effect: material from the first study period has a slight encoding advantage. CLT: novel complex material needs the freshest cognitive resources.
Block 2 (~Midday) 11:25–1:05pm	Application of Block 1 material; DCSPrep scenarios; retrieval practice; runbook writing; building tasks. Avoid: brand-new complex topic for the first time.	Moderate alertness. Applying and producing consolidates Block 1 encoding. This is the best interleaving window — connect Block 1 content to DCSPrep or Cisco material.
Block 3 (Afternoon, Post-Dip) 1:45–4:20pm	Hands-on application; DCSPrep app building; ticket/evidence writing; flashcard review; soft-skill roleplay; end-of-day evidence logging. Avoid: new abstract video content as primary activity.	Post-prandial dip 1:45–3:15pm. Active and applied tasks sustain engagement better than passive video. Recency effect: material from the last study period of the day also benefits from memory consolidation overnight.

Primacy and Recency Across a Day

The serial position effect (Murdock, 1962) — better recall for material at the beginning and end of a study sequence — applies within blocks and, more weakly, across a day. Material encoded in Block 1 and material encoded at the close of Block 3 both benefit from relative advantage. Block 2 material is most at risk of the 'muddy middle' effect and should therefore be the most application-heavy (application strengthens encoding even when initial encoding conditions are less optimal).

Thursday vs Friday: structuring them differently

Thursday is the primary **intake and encoding** day. Friday is the primary **retrieval, application, and consolidation** day. This means Friday's Block 1 brain-dump is the most important single activity of the week for retention of new material. Friday's Block 3 is the best window for Evidence Pack work and structured output, because by then you have had a full day to process and connect Thursday's content.

With the upcoming weekend providing 48 hours of additional consolidation after Friday, any material reviewed in Friday's blocks will receive two overnight sleep cycles before the following Thursday. Friday is therefore the optimal day for introducing material you want to be well-consolidated by next week — not necessarily brand-new complex material (reserve that for Thursday morning), but review-and-apply of the week's themes.

Sequencing Across the Week & Integrating Sources

Two PD days, five operational days, and multiple overlapping sources

The Five-Day Gap

The five days between Friday and the following Thursday function as the second, longer spacing interval. Research on spacing shows that longer intervals between practice sessions produce better long-term retention when the material has been adequately encoded in the first place (Cepeda et al., 2006; Smith & Scarf, 2017). The operational work during those five days provides incidental spaced retrieval — every time you look up a port number, escalate a ticket, or explain a network concept to a teacher, you are practising retrieval from Thursday/Friday's learning. This is not accidental; it is the mechanism through which professional learning consolidates into professional competence.

Deliberate low-effort incidental review during the gap: a 5-minute DCSPrep flashcard session on Tuesday or Wednesday morning, or writing a ticket note that explicitly references a concept studied last week, extends the spacing chain without requiring a full study block.

Multiple Sources on the Same Topics — Rich Interleaving or Shallow Hopping?

You have at minimum five sources covering IP/networking: CompTIA Core 1 (done), Core 2 (in progress), Cisco Networking Academy, La Trobe academic content, Udemy subnetting, and DCSPrep. This is an asset, not a problem — but only with discipline.

Rule of Source Commitment: commit to one source as 'primary anchor' for each topic until you can explain it to a DCSPrep novice-level persona without notes. Then — and only then — layer a second source for depth or perspective. Compulsively switching sources before achieving this benchmark produces shallow breadth without durable encoding.

- **IP subnetting:** Block in the Udemy course first (one or two full sessions). Then revisit with a Cisco lab exercise. Then write a DCSPrep explainer. Three sources, three encoding modes — but sequentially, not simultaneously.
- **Windows OS:** Messer Core 2 is the primary anchor (you are here now). Microsoft Learn can add depth for Entra/Intune specifics after the Messer section is complete.
- **Cybersecurity:** Messer Core 2 security sections → Cisco Cybersecurity pathway → DCSPrep threat scenario. Each layer adds application depth to the same conceptual foundation.

Completing Core 2 Messer (~10 hours remaining)

At approximately 2.5–3 hours of video content per Block 1 (at 1.25–1.5× speed with pausing), the remaining 10 hours of Core 2 represents roughly 3–4 Block 1 sessions. At the current Thursday/Friday cadence, completion of viewing is achievable within 3–4 weeks. However, **watching is not studying** — each hour of Core 2 video should produce roughly 20–30 minutes of retrieval practice and 20 minutes of application or writing. Budget 2 hours of follow-on work per hour of new video across the following Block 2 and Block 3 sessions.

Assessment Days & Deload Weeks

Every 4–6 weeks, allocate one full Friday to pure retrieval assessment: CompTIA practice exam (timed, full exam conditions), review wrong answers with written explanations from memory, and update the DCSPrep Evidence Pack with the result. This serves as both a metacognitive check and a manager-facing progress marker. A deload week — defined as a week with no new intake, only retrieval practice and Evidence Pack work — is protective after any week with two or more heavily interrupted blocks.

Ideal Mix of PD Activity Types

Evidence-based recommended balance across 13+ weekly PD hours

There is limited direct research on optimal activity mix ratios for IT certification preparation specifically. The recommendations below synthesise cognitive psychology principles (testing effect, generation effect, interleaving) with workplace learning research and the specific demands of an L1/L2 IT support role. They are directional, not algorithmic.

Activity Type	% of Weekly Time	Hours/Week	Evidence Rationale
Video Intake (Messer, Microsoft Learn, Cisco)	20–25%	2.7–3.3 hrs	Necessary for new schema building; diminishing returns if used for review. Watch once at 1.25×–1.5× speed, dual-coding (sketch while watching). CLT: worked example for novice topics.
Retrieval Practice (flashcards, free recall, self-quiz)	20–25%	2.7–3.3 hrs	Highest ROI per minute for long-term retention. Distribute across all blocks. FSRS-scheduled cards most efficient. Testing effect: $d \approx 0.55$.
Hands-on Application (real tickets, lab, scenario)	15–20%	2.0–2.7 hrs	Procedural knowledge consolidation. Applied transfer. Best in Block 2 and Block 3. Real DCS tickets count — document them in PD log.
Writing & Explanation (runbooks, ticket notes, Feynman explains)	15–20%	2.0–2.7 hrs	Generation effect (Bertsch et al., 2007). Self-explanation $g=0.55$. Every minute of runbook writing is a study act.
Building (DCSPrep app & content)	10–15%	1.3–2.0 hrs	Generation + application + portfolio evidence. Protege effect when writing scenarios for a novice persona. Sustainable in Block 3 dip hours.
Evidence Pack & PD Log	5–8%	0.7–1.1 hrs	Metacognitive scaffolding. Manager-facing progress visibility. Protective against burnout (perceived progress, autonomy).
Roleplay & Soft Skills (de-escalation, parent comms)	3–5%	0.4–0.7 hrs	Important but often neglected. At least one 30-min session per fortnight. Scenario-based roleplay is higher transfer than conceptual reading.
Reading & Note-Taking	3–5%	0.4–0.7 hrs	Lower ROI than retrieval unless combined with immediate recall. Use only when building initial understanding of a new concept before video.

How the mix should shift over time

- **Current phase (Core 2 intake + new topics):** weight toward video intake (25%) and immediate retrieval practice after each video session. Building and writing serve as depth rather than breadth.
- **Post-Core-2 completion (consolidation phase):** shift video time to zero for CompTIA; increase retrieval practice and full practice exam to 30–35%. Add Microsoft Learn and Cisco for depth on DCS-relevant topics.
- **Heavily interrupted week (one or more full blocks lost):** drop intake to zero. Retrieval practice only — 15–20 minute flashcard sessions per surviving block. No new content until a full block is available again.

Interruption Handling

The schedule must degrade gracefully — live support always wins

Gloria Mark's research (UC Irvine) found that knowledge workers take an average of 23 minutes 15 seconds to fully regain focus after a significant interruption. For ADHD learners, the attentional set-shifting cost is higher still. The goal is not to prevent interruptions — it is to minimise re-entry cost and prevent compound loss.

Scenario	At Interruption	At Re-entry	End of Block
5-min walk-up (teacher at door)	Note current position in 1 sentence on paper/app before leaving. Handle walk-up.	Re-read the 1-sentence note. Do not restart from the beginning of the section. Resume at the exact point noted. If flashcards: resume the next card.	Continue as planned. Log the interruption as 1 line in PD log. No assessment of loss needed.
20-min ticket (consumes mid-block time)	Save state: write 'Last action: _____. Next action: ____' (2 sentences). Open ticket. Handle fully.	Read the 2-sentence state note. Take 90 seconds to reconstruct mental context (re-prime). Resume the stated next action.	If >40 min of block remains: continue planned activity. If <40 min: switch to flashcard retrieval — works in short bursts without requiring long setup.
Half a block consumed (sustained support task)	Log the topic you were studying before interruption. Focus fully on support task.	Do not attempt to continue complex new intake. Use remaining time for retrieval practice or PD log writing — both are valuable in <45 min windows.	Log the interruption. Write a 1-line summary of the support task you handled (counts as applied learning documentation). Mark the planned activity as 'carry to next block'.
Full block consumed (major incident)	The block is gone. No partial study during a major incident — split attention produces both poor support outcomes and poor learning.	N/A — no PD time remains.	Log the incident in the PD log as an applied learning event. Write 3 sentences: what the problem was, what you did, what you'd do differently. This IS metacognitive professional development. No shame attached to a lost block.
Paul / DCS staff task — unknown duration	Same as above: note your study state in 2 sentences. Then give the task full attention.	Return to state note. If >30 min remaining: resume. If <30 min: retrieval practice only — no complex new intake.	If task was complex: log it. If block was largely lost: mark it as an interruption week, reduce expectations for remaining blocks accordingly.

The Re-entry Ritual (for ADHD context-switching cost)

The re-entry ritual is the single most protective design element for ADHD learners. It must be short (90 seconds), concrete, and non-optional:

- Read your 2-sentence state note
- Say aloud (or write): 'I was doing ____ and my next action is ____'
- Set a 25-minute timer to anchor the re-entry focus window
- Do NOT restart from the beginning of the topic — trust the note

App Design Implications

What a prescriptive PD scheduler needs to do — and what it must avoid

Inputs the App Needs

- Current block length remaining (derived from block start time + current time)
- Time of day (to select block type and appropriate activity)
- Self-reported energy/state (3-level: Low / Moderate / High — not a 10-point scale)
- Days since last review of each major topic (feeds spacing algorithm)
- Recent quiz performance per topic (feeds FSRS difficulty/stability model)
- Activity type used in the previous 30 minutes (to ensure rotation, not monotony)
- Interruption count this block (adjusts ambition downward if >2 interruptions)
- PD activity mix in current week (surfaces if intake is crowding out retrieval)
- Whether a deload week has been flagged (suppresses new intake suggestions)

Algorithm Ideas with Research Backing

- **FSRS for flashcard scheduling:** FSRS (Ye et al., 2022–2023; Anki 23.10) outperforms SM-2 by 20–30% fewer reviews for equivalent retention. Integrate FSRS-based scheduling natively into DCSPrep flashcards. Each card tracks difficulty, stability, and retrievability.
- **Topic interleaving enforcer:** if the last two activities were on the same topic, the next recommendation must be a different topic or modality. Research: interleaving produces 30–40% better transfer than blocking (Rohrer, 2012).
- **Activity rotation (modality manager):** cycle video → retrieval → application → writing within a single block. No activity type should occupy >40% of any block.
- **Post-prandial dip guard:** between 1:45 and 3:15pm, suppress passive video as the primary recommendation; surface application and building tasks instead.
- **Fatigue estimator:** at minute 80 of any 155-minute block with no logged break, surface a mandatory movement prompt. Do not allow dismissal without a 5-minute timer.
- **Interruption-aware scaling:** if 2+ interruptions have been logged in the current block, automatically downshift to retrieval-only mode for the remaining time.
- **Deload detector:** if 2+ full blocks have been lost to operational work in the same week, offer a 'consolidation mode' for remaining blocks — retrieval practice and logging only.
- **Weekly mix monitor:** display a simple weekly pie chart (no streaks, no gamification) showing hours by activity type. Flag if intake >40% of week's total time.

What the App Must NOT Do — Failure Modes

The most common failure modes in productivity and learning apps, drawn from self-determination theory (Ryan & Deci, 2000) and the gamification literature (Mekler et al., 2017):

- **Streak mechanics:** streak-loss aversion penalises recovery weeks and interruptions — exactly the weeks where psychological protection is most needed. No streaks.

- **Over-prescription:** prescribing activity down to the minute for an entire day erodes autonomy and triggers reactance. Recommend the next 25–30 minutes; let Josh choose the next one.
- **Gamified points for compliance:** extrinsic rewards for following the schedule undermine intrinsic motivation for the learning itself (Deci et al., 1999 meta-analysis, $d = -0.68$ for externally rewarded tasks).
- **Notification spam:** each notification is a context switch. The app should surface recommendations at session transitions, not throughout the session.
- **Optimisation theatre:** displaying too many metrics (velocity, retention %, Ease factor per card) creates anxiety and turns metacognition into overhead. Surface three numbers: hours this week, topics due for review, one gap identified.
- **Punishing missed sessions:** if a block is lost, the app acknowledges it, logs it, and moves on. No 'you missed 3 days' guilt messaging.

Designing for ADHD Without Nagging

Key principles: predictability over surprise, opt-in prompts over opt-out, visual state over textual instruction, short writing fields over long forms. Specifically:

- Session start: one writing field — '3-sentence intention'. Pre-populated with the seed from last session.
- Midpoint break: a countdown timer that starts automatically at the pre-set midpoint. Visual only. One tap to dismiss after 7 minutes.
- Interruption button: one tap, captures state in 2 sentences, clears the screen. No form to fill.
- Re-entry: displays the 2-sentence state note full-screen until dismissed. Prevents 'where was I?' spiral.
- PD log: three fields, pre-labelled, 140-character limit each. Forces concise metacognitive closure.

Reference Tables

Block shapes, PD mix, and interruption handling at a glance

Table R1 — Block Length × Session Shape × Evidence Anchors

Block	Suggested Internal Shape (minutes)	Evidence Anchors
90 min (Fri Block 2)	Orientation (5) → Intake/Retrieval (25) → Application (25) → Writing (25) → Break (8) → Log (2)	Testing effect (Roediger & Karpicke, 2006); CLT worked examples; Generation effect (Bertsch, 2007)
100 min (Thu Block 2)	Orientation+Recall (5) → Primary Activity (30) → Application/Build (30) → Microbreak (5) → Interleaved Retrieval (25) → Log+Handoff (5)	Spacing (Cepeda et al., 2006); Interleaving (Rohrer, 2012); Albulescu microbreaks (2022)
145 min (Thu Block 1)	Orientation+Brain-dump (8) → New Intake A (37) → Application A (30) → Movement Break (8) → Intake B or Retrieval (27) → Building/Writing (25) → Retrieval Close (8) → Log (2)	Retrieval practice; CLT; Post-prandial timing; Generation effect; Albulescu mandatory break
155 min (Thu/Fri Block 3; Fri Block 1)	Orientation (8) → Active Application A (32) → Active Application B (35) → 10-min Mandatory Break (10) → SRS Flashcard Review (30) → Writing/Evidence (25) → Wind-down (10) → Log+Seed (5)	Post-prandial dip guard; Albulescu (>10 min break for high cognition); FSRS scheduling; Generation effect; Metacognitive closure

Table R2 — Recommended Weekly PD Activity Mix (13 hrs 20 min total)

Activity Type	% of Week	Hours/Week	Research Basis
Video Intake	20–25%	2.7–3.3 hrs	Schema building, worked examples (CLT)
Retrieval Practice	20–25%	2.7–3.3 hrs	Testing effect $d=0.55$ (Rowland, 2014)
Hands-on Application	15–20%	2.0–2.7 hrs	Procedural transfer, contextual learning
Writing & Explanation	15–20%	2.0–2.7 hrs	Generation effect, self-explanation $g=0.55$
DCSPrep Building	10–15%	1.3–2.0 hrs	Generation + portfolio evidence
Evidence Pack / Log	5–8%	0.7–1.1 hrs	Metacognitive scaffolding, autonomy support
Roleplay / Soft Skills	3–5%	0.4–0.7 hrs	Applied professional competence
Reading & Notes	3–5%	0.4–0.7 hrs	Initial schema only; low ROI for review

Table R3 — Interruption Handling (Summary)

Scenario	At Interruption	At Re-entry	End of Block
5-min walk-up	Note position (1 sentence)	Read note, resume exact point	Log 1 line; continue
20-min ticket	Write state (2 sentences), handle	Read note, 90-sec re-prime, resume	Continue if >40 min left; else flashcards
Half block consumed	Log study topic, focus on support	Retrieval or log writing only	Carry planned activity to next block

Scenario	At Interruption	At Re-entry	End of Block
Full block consumed	Block is gone — full support focus	N/A	Log incident as applied learning (3 sentences)
Paul task / unknown	Note state, give task full attention	Return to note; if <30 min → retrieval	Log if complex; flag interruption week if needed

Table R4 — Representative Thursday at DCS (Full Day)

Time	Activity	Research Rationale
8:30–8:38	Block 1 Open	Write intention. Extended brain-dump from last Friday (free recall, no notes). Key retrieval moment of the week.
8:38–9:15	New Intake A	Core 2 Messer — next 2 sections (e.g., 1.6–1.7 Windows Settings). Dual-code: sketch config diagrams. 1.25× speed.
9:15–9:45	Application A	DCSPrep scenario or hands-on Windows 10 task based on 1.6–1.7 content. Reinforces just-watched schema.
9:45–9:53	Midpoint Break	Stand, walk around building. Brief outdoor if possible. Albulescu mandatory break — non-negotiable.
9:53–10:30	Intake B or Retrieval	If energy high: next Messer segment (1.8–1.9). If energy lower: SRS flashcard deck on Core 1 topics (spaced retrieval on older material).
10:30–10:53	Building / Writing	Write a runbook entry or DCSPrep scenario from memory on today's topic. Generation effect.
10:53–10:55	Log Close	3-sentence PD log. One gap noted. Seed for Block 2 written.
10:55–11:25	RECESS	Walk outdoors (15 min). Eat. No study content. Consolidation window.
11:25–11:30	Block 2 Open	Re-read Block 1 intention seed. Write new 3-sentence Block 2 intention.
11:30–12:00	Application of Block 1	DCSPrep Windows scenario or Microsoft Learn exercise on content from Block 1. Not more video.
12:00–12:30	Interleaved Retrieval	Flashcard deck: 20 cards from a DIFFERENT topic (e.g., networking or security). Deliberate interleaving.
12:30–13:00	Writing / Building	Ticket template writing from memory, OR DCSPrep content authoring. Protege framing.
13:00–13:05	Log Close	3-sentence log. Seed for Block 3.
13:05–13:45	LUNCH	Walk outdoors (20 min). Light protein-moderate meal. Sunlight exposure. No study content. Post-prandial dip preparation.
13:45–13:53	Block 3 Open	Active start — NOT passive video. Re-read lunch seed. Begin with a DCSPrep scenario or hands-on task.
13:53–14:25	Active Application	Troubleshooting scenario, IT lab, or hands-on task. High engagement to counteract dip.
14:25–15:00	DCSPrep Building	App features or scenario writing. Creative and applied — sustainable in dip window.
15:00–15:10	Mandatory Break	10-minute break. Move, hydrate, brief outdoor. Non-dismissible in the app.
15:10–15:40	Retrieval Practice	Full SRS deck review (20 cards). Interleaved topics. Low setup cost, high retention value.
15:40–16:10	Evidence Pack	Manager-safe documentation, Evidence Pack entries, ticket template finalisation.
16:10–16:18	Wind-down	Low-demand review of day's notes. Correct obvious gaps.

Time	Activity	Research Rationale
16:18–16:20	Log Close	Final 3-sentence log. Friday brain-dump seed written.

Table R4b — Same Thursday with Two Interruptions (Degraded Version)

Interruption 1: A teacher arrives at 9:25am (Block 1, during Application A) for a 20-minute fix. Interruption 2: Paul assigns a deployment task at 11:50am (Block 2, mid-retrieval) consuming 35 minutes.

Time	Activity	Rationale
8:30–9:25	Block 1 — as planned	No change. Full 55 minutes before interruption.
9:25	INTERRUPTION 1	Write 2-sentence state note. Handle teacher fix (20 min).
9:45	Re-entry	Read state note. 90-second re-prime. Resume Application A from where noted.
9:45–10:30	Application A + Intake B	Complete application. If time: 15 min further retrieval or Messer segment. Scale down: one activity, not two.
10:30–10:55	Building/Log	Shorten building to 15 minutes. Full log closure still happens (non-negotiable 3 min).
10:55–11:25	RECESS	No change. Walk, eat.
11:25–11:50	Block 2 — partial	25 minutes of planned retrieval before interruption 2.
11:50	INTERRUPTION 2	Write 2-sentence state note. Handle Paul's deployment task.
12:25	Re-entry	35 minutes of block remain. Switch to flashcard retrieval only — no complex new intake.
12:25–13:00	Flashcard Retrieval + Log	20 cards SRS deck. 3-sentence log. Mark planned building as 'carry to Block 3'.
13:05–13:45	LUNCH	No change. Walk, eat, sunlight.
13:45–16:20	Block 3 — as planned	Block 3 proceeds as standard. Carry the building/Evidence Pack work from Block 2 into the Block 3 writing window.

Watch-Outs

Patterns that look productive but the evidence suggests are not

■ Watching a video twice instead of testing yourself once

Re-watching produces familiarity, not recall. After the first viewing, your brain recognises the content easily — and mistakes that recognition for having learned it. Test instead.

■ Treating 'I understand it' as the same as 'I can retrieve and apply it'

Comprehension during a Messer video is a baseline, not an endpoint. The gap between 'it makes sense' and 'I can explain it to a teacher without notes on Tuesday' is where most learning effort lives.

■ Switching sources whenever a topic feels hard

Source-switching is often a form of avoidance. A topic that still feels unclear after Messer is usually better served by a DCSPrep scenario on that topic — not by finding a different video explanation.

■ Using breaks for 'quick' content consumption

A podcast during the recess walk depletes the same attentional system that was just working. Quiet walking is more restorative than informative walking.

■ Treating interrupted blocks as failed blocks

A block with two interruptions and 45 productive minutes of retrieval practice is a better week than a block you skipped entirely due to frustration. The compound cost of zero is higher than the compound cost of reduced.

■ The strict 25/5 Pomodoro as an evidence-based rule

The specific 25/5 numbers have no peer-reviewed basis as optimal cognitive work intervals. Use them if they suit you, but don't let the timer dictate your block structure. A 37-minute deep-work segment that matches the natural end of a Messer section is better than cutting it at 25.

■ Learning styles matching (VAK)

Not supported by the research (Pashler et al., 2008). Modality variation is valid for engagement and fatigue management — not because your 'visual brain' learns differently than your 'auditory brain'.

■ Gamified streaks in learning apps

Streak mechanics punish the weeks when you most need protection — heavily interrupted weeks and recovery weeks. Any streak-based system should be removed or turned off.

■ Confusing building DCSPrep with avoiding study

Building DCSPrep IS study, via the generation and protege effects. The risk is building features or UI polish that does not require encoding new content. Ensure that at least 50% of DCSPrep build time involves writing new scenarios, questions, or explanations from memory.

Sustaining This Rhythm for Years

A final reflection on the long game

Most learning research is conducted over weeks or months, with undergraduates in controlled settings. The honest caveat is that the evidence base for multi-year self-directed professional learning in a school IT support role is thin. The principles generalise, but the implementation must adapt to what life actually looks like in Year 2 and Year 3, not just Week 1.

Visible progress is the greatest protector against burnout

Your Evidence Pack is not administrative overhead — it is the mechanism by which you can see that you are better at your job than you were six months ago. Update it weekly, even briefly. The moment it feels like pointless paperwork is a signal to reconnect it to a real scenario where the skill mattered.

The role is the curriculum

Every ticket you resolve is a spacing event for something you studied. Every escalation you write is a generation act. The five days between PD blocks are not a gap in your learning — they are the extended spacing interval where operational practice consolidates what the blocks encode. Trust the architecture.

Deload weeks are not failure weeks

A fortnight with two heavily interrupted blocks and minimal new intake followed by a full-practice retrieval week is an excellent learning pattern. The rhythm is not a treadmill — it's a wave. If the wave flattens for two weeks, it will crest again. Log it, acknowledge it, and return.

The app serves the person, not the other way round

When the app feels like a supervisor rather than a scaffold, it has overstepped. The prescriptive function should recommend, not require. You are the professional; the app is the prompting system. If a recommendation doesn't fit the moment, override it without guilt — and consider whether the recommendation logic needs updating.

The endpoint is not the A+ certificate

The CompTIA certification is a milestone on a longer arc toward full L2 capability, confident teacher/student support, and recognised expertise at DCS. The rhythm you build now — spacing, retrieval, generating, applying, recovering — is transferable to any domain you encounter next: Entra ID, Intune, Azure, whatever comes after. You are not just studying CompTIA content; you are building a learning architecture that will serve you for decades.

This brief was compiled from peer-reviewed primary literature and meta-analyses. Where strong evidence is absent, this has been stated explicitly. Key sources include: Cepeda et al. (2006) *Psychological Bulletin*; Burgess & Jilbert (2024) *Behavioral Sciences*; Rowland (2014) *Psychological Bulletin*; Parker & Roessger (2020) *IJLE*; Bjork & Bjork (2020); Albulescu et al. (2022) *PLOS ONE*; Sweller (1988, 2024); Kalyuga et al. (2003); Bertsch et al. (2007); Fiorella & Mayer (2016); Pashler et al. (2008) *PSPI*; Ye et al. *FSRS (2022–2023)*; Guttesen et al. (2025) *Cognition*; Mark et al. *UC Irvine (interruption research)*; Monk (2005) *post-prandial dip*; Erickson et al. (2011) *PNAS*; Kaplan & Kaplan (1989) *ART*.

End of DCS PD Research Brief · Compiled 08 May 2026